

Exponent Laws with Variable Bases

MA.8.AR.1.1

I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

Yesterday the bases were all numbers.

Today the base is a letter. Every law still works.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$
2. Write as one power: $\frac{6^7}{6^5}$
3. $\sqrt{50}$ is between which two whole numbers?
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$

Warm-Up

Answers.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$ **4**
2. Write as one power: $\frac{6^7}{6^5}$ **$6^2 = 36$**
3. $\sqrt{50}$ is between which two whole numbers? **7 and 8**
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$ **y^5**

Notes

Copy both lines.

I. Product of powers — the base is a letter now

$$x^2 \cdot x^3 = (x \cdot x)(x \cdot x \cdot x) = x^5$$

$$x^m \cdot x^n = x^{m+n}$$

$$3x^2 \cdot 5x^4 = (3 \cdot 5)(x^2 \cdot x^4) = 15x^6$$

Multiply the coefficients. Add the exponents of like bases.

Different variables stay separate: $x^2y \cdot x^3y^4 = x^5y^5$

Notes

II. Power of a power, power of a product

$$(x^3)^4 = x^3 \cdot x^3 \cdot x^3 \cdot x^3 = x^{12}$$

$$(x^m)^n = x^{mn}$$

$$(2y^2)^3 = 2^3 \cdot (y^2)^3 = 8y^6$$

$$(15xy^2)^3 = 3,375x^3y^6$$

Power of a power: multiply the exponents. Power of a product: raise every factor.

Notes

III. Quotient of powers — dividing monomials

$$\frac{x^7}{x^2} = \frac{x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x}{x \cdot x} = x^5$$

$$\frac{x^m}{x^n} = x^{m-n}$$

$$\frac{12a^5}{4a^2} = 3a^3$$

$$\frac{m^6n^4}{m^2n^4} = m^4n^0 = m^4$$

Divide the coefficients. Subtract the exponents of like bases.

Example 1

Multiplying two monomials.

$$(4x^3y)(-2x^2y^5)$$

Write an equivalent expression as a single monomial.

Example 1

Worked.

$$(4x^3y)(-2x^2y^5) = (4)(-2) \cdot x^3x^2 \cdot y^1y^5 \quad \text{regroup: coefficients, } x\text{'s, } y\text{'s}$$
$$= -8 \cdot x^5 \cdot y^6 = -8x^5y^6 \quad \text{multiply coefficients; add exponents}$$

Answer: $-8x^5y^6$

Your Turn

Same steps, your numbers. Boards up when done.

$$(5a^2b^4)(3ab^2)$$

Your Turn

Answer.

$$(5a^2b^4)(3ab^2)$$

$$5 \cdot 3, a^2 \cdot a^1, b^4 \cdot b^2$$

$$\textbf{Answer: } 15a^3b^6$$

Example 2

A power of a product, then a quotient.

$$\frac{(3m^2n)^3}{9m^4n}$$

Write an equivalent expression as a single monomial.

Example 2

Worked.

$$(3m^2n)^3 = 27m^6n^3 \quad \text{power of a product: raise all three factors}$$

$$\frac{27m^6n^3}{9m^4n} = 3 \cdot m^{6-4} \cdot n^{3-1} = 3m^2n^2 \quad \text{divide coefficients; subtract exponents}$$

Answer: $3m^2n^2$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(2p^3)^4}{8p^5}$$

Your Turn

Answer.

$$\frac{(2p^3)^4}{8p^5}$$

$$16p^{12} \text{ over } 8p^5$$

Answer: $2p^7$

Whiteboards • Question 1 of 9

Boards up on three.

$$x^5 \cdot x^8$$

Answer it.

Write as a single power.

Whiteboards • Question 1 of 9

Answer.

$$x^5 \cdot x^8$$

like bases: add the exponents

Answer: x^{13}

Whiteboards • Question 2 of 9

Boards up on three.

$$(y^4)^3$$

Answer it.

Write as a single power.

Whiteboards • Question 2 of 9

Answer.

$$(y^4)^3$$

power of a power: multiply

Answer: y^{12}

Whiteboards • Question 3 of 9

Boards up on three.

$$(2x^3)^4$$

Answer it.

Write as a single monomial.

Whiteboards • Question 3 of 9

Answer.

$$(2x^3)^4$$

raise the 2 as well: $2^4 \cdot x^{12}$

Answer: $16x^{12}$

Whiteboards • Question 4 of 9

Boards up on three.

$$(3a^2b)(4ab^5)$$

Answer it.

Write as a single monomial.

Whiteboards • Question 4 of 9

Answer.

$$(3a^2b)(4ab^5)$$

$$3 \cdot 4, a^2 \cdot a, b \cdot b^5$$

$$\textbf{Answer: } 12a^3b^6$$

Whiteboards • Question 5 of 9

Boards up on three.

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer: A — $25m^6$

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^6$

B. $4x^4$

C. $9x^6$

D. $4x^{10}$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^6$

B. $4x^4$

C. $9x^6$

D. $4x^{10}$

Answer: A — $4x^6$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{m^6 n^4}{m^2 n^4}$$

Answer it.

Write as a single monomial.

Whiteboards • Question 7 of 9

Answer.

$$\frac{m^6 n^4}{m^2 n^4}$$

$$m^4 \cdot n^0, \text{ and } n^0 = 1$$

Answer: m^4

Whiteboards • Question 8 of 9

Boards up on three.

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

Answer it.

Which law applies?

Whiteboards • Question 8 of 9

Answer.

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

$$3 + n = 11$$

Answer: $n = 8$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

Write your answer in sentences.

Say what each of them did.

Whiteboards • Question 9 of 9

Answer.

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

both — Justina's is Rachel's product, multiplied out

Answer: Both are correct; $15^3 = 3,375$, $x \cdot x \cdot x = x^3$, and six y 's make y^6 .

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Multiply powers: variable bases — TR9

Powers of a power: variable bases — 2CU

Divide monomials with positive exponents — C8L